

CLASS 5 MATH

Advanced Single-Column Workbook

BODMAS, Fractions, Geometry & 100 Olympiad Puzzles

Chapter 1: Millions and Crores

SECTION A: CORE CONCEPTS

1. Write the number 45,067,890 in words using the International System.

2. Write the number 3,45,67,890 in words using the Indian Numeral System.

3. What is the difference between the place value and face value of 7 in 8,765,432?

4. Round 4,567,892 to the nearest million.

5. Arrange in descending order: 345,678; 354,678; 345,876; 354,876.

SECTION B: REAL WORLD APPLICATIONS

1. The population of California is approximately 39,000,000. Write this number in expanded form.

2. Crunchley sold 1,250,450 packets of makhana in Q1 and 1,305,000 in Q2. How many total packets were sold? Write the final answer in words.

3. Atharv's coding project, Jigyasa, processed 8,456,200 voice requests. Round this number to the nearest hundred thousand.

Chapter 2: Operations & BODMAS

SECTION A: CORE CONCEPTS

1. Solve using BODMAS: $45 + 15 \div 3 \times 2 - 10$.

2. Multiply: 4,567 by 234.

3. Divide 85,674 by 14. Find the quotient and remainder.

4. Simplify: $100 - [50 - \{ 30 - (10 + 5) \}]$.

5. Find the product of the largest 3-digit number and the smallest 2-digit number.

SECTION B: REAL WORLD APPLICATIONS

1. A school ordered 1,450 books for the library in Milpitas. If each book costs \$15, what is the total amount paid?

2. Priyanka has 12,450 cherries from the orchard. She packs them into boxes of 25. How many boxes will she fill, and how many cherries are left over?

3. Mukesh bought an EGO Power+ lawn mower for \$450 and a cultivator for \$150. If he pays in 12 equal monthly installments, how much will he pay each month?

Chapter 3: Fractions, Decimals, & Percentages

SECTION A: CORE CONCEPTS

1. Add the mixed fractions: $3\frac{1}{2} + 4\frac{2}{3}$.

2. Multiply: $\frac{4}{5} \times \frac{15}{16}$.

3. Divide: $\frac{7}{8} \div \frac{3}{4}$.

4. Convert the fraction $\frac{4}{5}$ into a decimal and a percentage.

5. What is 35% of 600?

SECTION B: REAL WORLD APPLICATIONS

1. Atharv completed $\frac{3}{8}$ of his math worksheet before lunch and $\frac{1}{4}$ after lunch. What fraction of the worksheet is left to complete?

2. Anaya has a ribbon that is 15.75 meters long. She cuts it into 5 equal pieces. How long is each piece?

3. A store offers a 20% discount on a \$150 toy set. How much will Priyanka pay after the discount?

Chapter 4: Geometry, Volume & Time

SECTION A: CORE CONCEPTS

1. Find the volume of a cube with a side length of 8 cm.

2. Find the perimeter and area of a rectangle with length 15m and width 12m.

3. Convert 450 minutes into hours and minutes.

4. What is the supplementary angle of 75 degrees?

5. Name a triangle that has all three sides of different lengths.

SECTION B: REAL WORLD APPLICATIONS

1. A rectangular shipping box for Crunchley makhana is 40cm long, 30cm wide, and 20cm high. What is the volume of the box?

2. Atharv's soccer game in Sunnyvale started at 10:15 AM and ended at 11:45 AM. How many minutes did the game last?

3. A square garden needs fencing. If one side of the garden is 25 meters, and the fencing costs \$8 per meter, what is the total cost?

Multi-Concept Combined Practice

Level 1: Core Proficiency (Average Student) - SET 1

1. Write $5,000,000 + 400,000 + 30,000 + 200 + 10 + 5$ in standard form.

2. Add the decimal numbers: $2.50 + 3.75$.

3. Subtract the fraction $\frac{3}{8}$ from 1 whole.

4. Find the perimeter of a rectangle with a length of 15 cm and a width of 10 cm.

5. What is 50% of 200?

6. Divide 456 by 12. Show your steps.

7. Round the number 45,678 to the nearest thousand.

8. A train leaves San Jose at 2:30 PM and arrives in Los Angeles at 8:15 PM.
How long was the journey?

9. Find the Least Common Multiple (LCM) of 6 and 8.

10. Write 0.75 as a fraction in its simplest form.

Multi-Concept Combined Practice

Level 1: Core Proficiency (Average Student) - SET 2

1. Write 12,345,678 in words using the International System.

2. Multiply 245 by 36.

3. Add the fractions: $\frac{1}{4} + \frac{1}{2}$.

4. Find the area of a square with a side length of 12 meters.

5. Convert 3.5 kilograms to grams.

6. Find the Highest Common Factor (HCF) of 12 and 18.

7. What is 25% of 400?

8. A box holds 24 packets of Crunchley makhana. How many packets are in 15 boxes?

9. Solve using BODMAS/Order of Operations: $15 + 5 \times 4$.

10. Which fraction is greater: $\frac{5}{6}$ or $\frac{7}{8}$? Show your working.

Multi-Concept Combined Practice

Level 2: Advanced Application (Top 10%) - SET 1

1. Priyanka bought a shirt for \$25.50 and pants for \$45.75. She paid with a \$100 bill. What is her exact change?

2. A rectangular swimming pool is 20m long, 10m wide, and 2m deep. What is its volume in cubic meters?

3. Solve using BODMAS: $50 - (10 + 5) \times 2$.

4. The LCM of two numbers is 36, and their HCF is 6. If one number is 12, what is the other number?

5. Write the fraction $\frac{5}{8}$ as a decimal.

6. If $\frac{3}{4}$ of a specific number is 60, what is the original number?

7. Atharv scored 85, 90, and 95 on his first three math tests. What is his average score?

8. Find the missing number in the sequence: 2, 5, 10, 17, 26, ____.

9. Anaya is 10 years old. Her father Mukesh is 4 times as old. In how many years will her father be exactly twice her age?

10. A recipe for cookies calls for 1.5 cups of flour. How many cups of flour are needed to make 4 batches?

Multi-Concept Combined Practice

Level 2: Advanced Application (Top 10%) - SET 2

1. A car travels 240 miles in 4 hours. What is its average speed in miles per hour?

2. Add the mixed numbers: $2\frac{1}{2} + 3\frac{3}{4}$.

3. Find the cost of fencing a square garden with a side length of 25m at the rate of \$5 per meter.

4. Convert 4 hours and 45 minutes completely into minutes.

5. What is the value of $3^3 + 4^2$?

6. A water tank is $\frac{3}{5}$ full. It needs 40 more liters to be completely full. What is its total capacity?

7. Solve the decimal multiplication: 12.5×0.4 .

8. The sum of two consecutive even numbers is 54. What are the two numbers?

9. Find the prime factorization of 72.

10. How many lines of symmetry does a regular hexagon have?

Multi-Concept Combined Practice

Level 3: Logical Reasoning (Top 1%) - SET 1

1. 5 workers can build a wall in 10 days. How many days will it take for 10 workers to build the exact same wall?

2. A clock shows exactly 3:15. What is the exact angle in degrees between the hour and minute hands? (Careful, the hour hand moves!).

3. Find the sum of all prime numbers between 20 and 30.

4. A train 150 meters long crosses a stationary pole in 10 seconds. What is the train's speed in meters per second?

5. A vendor bought 100 apples from an orchard for \$50 and sold them at \$0.75 each. What is his profit percentage?

6. How many diagonals does a regular octagon have?

7. If $x + y = 15$ and $x - y = 5$, find the value of x multiplied by y .

8. What is the units digit (last digit) of 7 to the power of 25?

9. Anaya has \$2.50 in quarters (\$0.25) and dimes (\$0.10). She has 16 coins in total. How many quarters does she have?

10. Fill in the missing number in the sequence: 1, 8, 27, 64, ____.

Multi-Concept Combined Practice

Level 3: Logical Reasoning (Top 1%) - SET 2

1. The ratio of boys to girls in a class is 3:2. If there are 30 students in total, how many are girls?

2. A book has 300 pages. How many times is the digit '2' printed in the page numbers from 1 to 300?

3. What is the remainder when the number 1,234,567 is divided by 9? (Use divisibility rules).

4. A rectangle has a perimeter of 40cm. Its length is 3 times its width. Find the total area of the rectangle.

5. If 20% of a number is 45, what is 50% of that same number?

6. Simplify the complex fraction: $(\frac{1}{2} + \frac{1}{3}) \div (\frac{1}{4})$.

7. If 4 cats can catch 4 mice in 4 minutes, how many cats are needed to catch 100 mice in exactly 100 minutes?

8. Find the sum of the interior angles of a hexagon.

9. A piece of wood 10 meters long is cut into 5 equal pieces. If each cut takes 2 minutes, how long does the whole job take?

10. What number is half of a quarter of a tenth of 800?

Multi-Concept Combined Practice

Level 4: Critical Thinking (Top 0.1%) - SET 1

1. I have a 3-liter jug and a 5-liter jug, and an unlimited supply of water. List the steps to measure exactly 4 liters.

2. Solve the Cryptarithm: $SEND + MORE = MONEY$. Every letter represents a unique digit from 0-9. What digit does 'M' represent?

3. If 5 machines make 5 widgets in 5 minutes, how long does it take 100 machines to make 100 widgets?

4. A solid cube with a side length of 4cm is painted red on all sides. It is then cut into smaller 1cm cubes. How many of the small cubes have exactly ONE red face?

5. How many triangles in total can you find in a regular star polygon (pentagram)?

6. A bat and a ball cost \$1.10 in total. The bat costs exactly \$1.00 more than the ball. How much does the ball cost?

7. There are 10 people in a room. Each person shakes hands with everyone else exactly once. How many total handshakes occur?

8. Find the missing term in the sequence: 2, 3, 5, 9, 17, ____.

9. A palindrome number reads the same backwards. Priyanka's car odometer reads 15951. How many miles must she drive to reach the next palindrome?

10. Two trains are 100 miles apart, moving towards each other at 50 mph. A fast bird flies back and forth between them at 75 mph until they meet. How far does the bird fly in total?

Multi-Concept Combined Practice

Level 4: Critical Thinking (Top 0.1%) - SET 2

1. You have 9 identical-looking coins. 8 weigh the same, but 1 is slightly heavier. Using a balance scale only twice, how do you find the heavy coin?

2. Find the smallest positive number that leaves a remainder of 1 when divided by 2, 3, 4, 5, and 6, but is perfectly divisible by 7.

3. Three friends check into a hotel room that costs \$30. They each pay \$10. The clerk realizes the room is \$25. He gives \$5 to the bellboy to return. The bellboy keeps \$2 and gives \$1 back to each friend. They paid \$9 each (\$27), plus \$2 for the bellboy = \$29. Where is the missing dollar? Explain the logical fallacy.

4. What is the last digit of 2 to the power of 2026?

5. A frog is at the bottom of a 30-foot well. Each day it jumps up 3 feet, but slips down 2 feet at night. How many days until it escapes the well?

6. Complete a 3x3 magic square using the numbers 1 through 9 so that every row, column, and diagonal adds up to exactly 15.

7. If you divide 30 by a half ($\frac{1}{2}$) and then add 10, what do you get?

8. A man has 53 socks in a dark drawer: 21 blue, 15 black, 17 red. How many must he pull out blindly to guarantee he has at least one matching pair?

9. A clock takes 5 seconds to strike 6 o'clock (6 chimes). How long will it take to strike 12 o'clock?

10. Evaluate the alternating series: $1 - 2 + 3 - 4 + 5 - 6 \dots + 99 - 100$.

Multi-Concept Combined Practice

Level 5: National Olympiad (Top 10) - SET 1

1. A farmer has some chickens and cows. He counts 50 heads and 140 legs in total. How many chickens does he have?

2. You are in a hallway with 100 closed lockers. You open every locker. Then you close every 2nd locker. Then you toggle every 3rd locker (open if closed, close if open), then every 4th, up to 100. Which specific lockers remain open at the end?

3. How many 3-digit numbers contain the digit '5' at least once?

4. Four people need to cross a rickety bridge at night. They have one flashlight. They cross at speeds of 1, 2, 5, and 10 minutes. Maximum 2 people can cross at a time, and they must carry the flashlight. What is the shortest possible time for all to cross?

5. A magical spider web doubles in size every day. If it takes 30 days to completely cover a window, on what day was the window exactly half covered?

6. What is the maximum number of pieces you can cut a circular pizza into using exactly 4 straight planar cuts?

7. If A, B, and C are distinct non-zero digits and $AAA + BBB + CCC = BAAC$, what are the values of A, B, and C?

8. Find the sum of all the factors of 360.

9. You have 12 bags of coins. One bag has fake coins that weigh 9g instead of the real 10g. You have a digital scale. How can you find the fake bag in just ONE weighing?

10. How many total rectangles (of any size) can you count in a 4x4 grid?

Multi-Concept Combined Practice

Level 5: National Olympiad (Top 10) - SET 2

1. The sum of the digits of a 2-digit number is 10. If 18 is added to the number, its digits are reversed. What is the original number?

2. A king has 1000 bottles of wine, but 1 is poisoned. The poison acts in exactly 24 hours. He has 10 prisoners available to taste the wine. How can he identify the poisoned bottle in just 24 hours?

3. Using exactly four 4s and any math operations (+, -, x, /, brackets, factorials), how do you make the number 20?

4. There is a monkey on a greased pole 60 feet high. He climbs 3 feet in one minute, but slips down 2 feet in the next minute. How many minutes will it take him to reach the top of the pole?

5. A hollow cube has an external volume of 1000 cubic cm and its walls are 1 cm thick. What is the volume of the empty space inside?

6. Two cars travel from City A to City B. Car 1 travels half the total DISTANCE at 30mph and half at 60mph. Car 2 travels half the total TIME at 30mph and half the TIME at 60mph. Which car arrives first?

7. Find the Greatest Common Divisor (GCD) of 111,111 and 11,111,111.

8. You have a 7-minute hourglass and an 11-minute hourglass. How do you boil an egg for exactly 15 minutes?

9. What day of the week will it be exactly 10^6 days from today, assuming today is Thursday?

10. There are 25 horses. You can race up to 5 at a time, but you don't have a stopwatch to record times, only their relative finishing positions. What is the minimum number of races needed to identify the top 3 fastest horses?
